

Monitoring And Automation System FOR LATHE MACHINES

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This pilot system offers an Arduino-based solution for real-time monitoring and automation of lathe machines, integrating temperature, vibration, and operational data to improve efficiency and safety.

Designed to monitor temperature and vibration in real time, the system enables automated alerts and preventive actions to minimise downtime and extend machine life. It also enhances operator convenience by displaying and transmitting data on an LCD via serial communication.

Vibrations are detected through analogue and digital signals, while temperature is measured using an LM35 sensor. Anomalies trigger alerts through a buzzer. Real-time data is shown on a 16x2 LCD display.

The system can be modified for expanded automation to include a relay module that activates external mechanisms. This approach supports predictive maintenance, ensures safe operating conditions, prevents damage, and enhances overall productivity. The prototype is shown in Fig. 1. Refer to the Bill of Materials table to build this system.

EFY note. Although the prototype shown uses an Arduino Uno, the circuit also works with an Arduino Nano, as shown in Fig. 3.

Circuit and working

The circuit diagram of the monitoring and automation system for lathe machines is shown in Fig. 2. It is built around an Arduino Nano board (MOD1), a vibration sensor (S1), a temperature sensor LM35 (S2), a

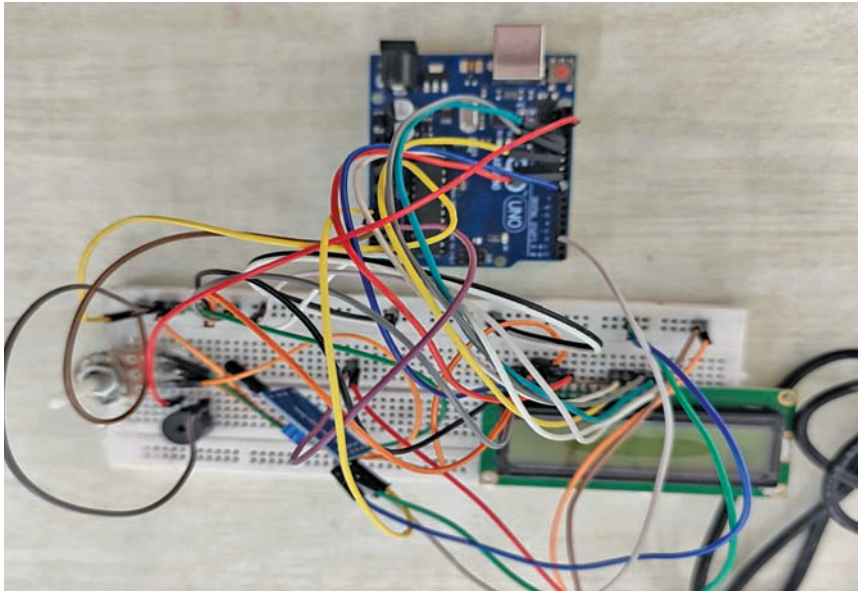


Fig. 1: Authors' prototype

buzzer, and other components.

Vibration monitoring. A vibration sensor connected to analogue pin A0 and digital pin D2 detects machine vibrations. Analogue readings are converted to voltage and scaled to vibration speed. The system alerts via the LCD and serial monitor if vibrations exceed a predefined threshold.

Temperature monitoring. The LM35 sensor connected to analogue pin A1 continuously measures the machine's temperature. The Arduino processes the analogue signal to determine the temperature in °C and °F. If the temperature exceeds 70°C, a buzzer is activated, and the LCD displays a warning.

A 16x2 LCD connected to digital pins D8-D13 displays real-time temperature and vibration data.

BILL OF MATERIALS

Components	Quantity
Arduino Nano (MOD1)	1
Potmeter 10k (POT1)	1
16x2 LCD display (MOD2)	1
Vibration sensor (S1)	1
LM35 temperature sensor (S2)	1
Buzzer (B1)	1
USB cable (for 5V)	1

The serial monitor provides a log of machine conditions for detailed analysis. The buzzer and relay offer immediate auditory and operational responses. The relay module can be configured to trigger external systems such as cooling fans or shutdown mechanisms in critical conditions.

EFY note. This system currently uses an SPI-based LCD, but